

A Mini Project Report on

## **IoT-based Voice-enabled Smart home Automation system**

Submitted for the partial fulfilment of the academic requirements for the  
Award of the Degree of

### **BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted By

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## **CERTIFICATE**

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This is to certify that the work reported in the present project entitled “**Iot-Based Voice-Enabled Smart Home Automation System**” is a record of work done by us in the Department of Electronics and Communication Engineering, Vidya Jyothi Institute of Technology, Hyderabad. The report is based on the project work done entirely by us and not copied from any other sources.

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## ABSTRACT

This project presents an **Iot-Based Voice-Enabled Smart Home Automation System** using the ESP32 microcontroller. The system is designed to automate and control home appliances based on environmental conditions and user commands. It integrates sensors such as DHT11 for temperature and humidity detection and an LDR sensor for light intensity monitoring.

The ESP32 acts as the central controller with built-in Wi-Fi and Bluetooth capabilities, enabling both remote and local control. Appliances such as lights and fans are controlled using relay modules. The system operates in both **automatic mode** (based on sensor readings) and **manual mode** (via web interface or Bluetooth commands).

The fan is activated when temperature exceeds a threshold, and lights are controlled based on ambient light conditions. Additionally, users can control devices through a web dashboard or voice commands via Bluetooth.

This system provides a **low-cost, efficient, and scalable solution** for smart home automation and demonstrates the practical implementation of IoT in real-world applications.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Overview of Smart Home Automation

Smart home automation is one of the most rapidly evolving applications of modern electronics and communication technologies. It refers to the use of intelligent systems and interconnected devices to monitor and control household appliances automatically, with minimal or no human intervention. The concept of smart homes has gained immense popularity due to its ability to enhance comfort, security, energy efficiency, and convenience in daily life.

In a traditional home setup, appliances such as lights, fans, air conditioners, and other electrical devices are manually controlled using switches. This approach, although simple, is inefficient in terms of energy usage and requires continuous human effort. Smart home systems eliminate these limitations by introducing automation, where devices can operate based on environmental conditions or user commands.

A typical smart home system consists of sensors, controllers, communication modules, and actuators. Sensors are used to collect real-time data from the environment, such as temperature, humidity, or light intensity. This data is processed by a controller, which then takes appropriate decisions to control appliances. Communication technologies such as Wi-Fi and Bluetooth enable remote monitoring and control of these systems.

With advancements in embedded systems and wireless communication, smart home automation has become more accessible and affordable. Microcontrollers like ESP32 have made it possible to develop compact and efficient automation systems that can be easily integrated into homes. These systems are not only used in residential buildings but also in offices, industries, and smart cities.

## **1.2 Importance of IoT in Home Automation**

The Internet of Things (IoT) plays a vital role in the development of modern smart home systems. IoT refers to a network of interconnected devices that can communicate with each other and exchange data over the internet. These devices are embedded with sensors, software, and communication capabilities, allowing them to operate intelligently and autonomously.

In the context of home automation, IoT enables seamless interaction between various devices such as sensors, controllers, and appliances. For example, a temperature sensor can detect changes in room temperature and send this information to a controller via the internet. The controller can then activate a fan or air conditioner automatically, without any manual intervention.

One of the key advantages of IoT-based systems is remote accessibility. Users can monitor and control their home appliances from anywhere in the world using smartphones or computers. This is particularly useful for managing energy consumption and ensuring safety. For instance, users can turn off appliances remotely if they forget to do so before leaving home.

IoT also enhances energy efficiency by enabling intelligent decision-making. Devices can operate only when necessary, reducing unnecessary power consumption. For example, lights can be turned off automatically during daylight hours using light sensors.

Another important feature of IoT is scalability. Additional devices and sensors can be easily integrated into the system without major modifications. This makes IoT-based home automation systems flexible and adaptable to future requirements.

Furthermore, IoT contributes to data-driven decision-making. By analyzing data collected from sensors, users can gain insights into energy usage patterns and optimize their consumption accordingly.

In summary, IoT serves as the backbone of modern smart home systems, enabling connectivity, automation, and intelligent control

### 1.3 ESP32 as a Smart Controller

The ESP32 microcontroller is widely used in IoT applications due to its powerful features and versatility. It is developed by Espressif Systems and is considered an advanced version of the ESP8266. The ESP32 integrates both Wi-Fi and Bluetooth capabilities, making it an ideal choice for wireless communication in smart home systems.

One of the major advantages of ESP32 is its ability to perform multiple functions simultaneously. It can read sensor data, process information, and communicate with other devices, all within a single chip. This reduces the need for additional hardware and simplifies system design.

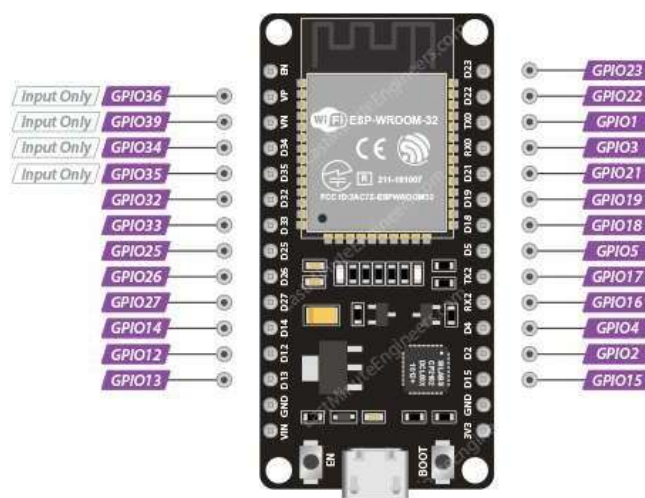
The ESP32 is equipped with a dual-core processor, which provides high processing speed and efficiency. It also supports multiple input/output (I/O) pins, allowing it to interface with various sensors and devices such as relays, motors, and displays.

Another important feature of ESP32 is its low power consumption. It includes different power-saving modes, making it suitable for energy-efficient applications. This is particularly important in IoT systems, where devices may need to operate continuously for long periods.

In this project, the ESP32 acts as the central controller that receives data from sensors such as the DHT11 and LDR. Based on the sensor readings, it processes the data and controls appliances through relay modules. Additionally, it hosts a web server and enables Bluetooth communication, allowing users to control the system manually.

Due to its affordability, compact size, and powerful capabilities, ESP32 has become one of the most popular microcontrollers for IoT-based smart home applications.

Fig 1.3 ES32 as a Smart controller



#### **1.4 Need for Smart Home Automation Systems**

With the rapid growth of technology and increasing demand for convenience, traditional home systems are no longer sufficient to meet modern requirements. There is a growing need for smart home automation systems that can provide intelligent control, energy efficiency, and enhanced security.

One of the primary reasons for adopting smart home systems is energy conservation. In conventional systems, appliances are often left ON unnecessarily, leading to wastage of electricity. Smart systems address this issue by automatically controlling devices based on environmental conditions, thereby reducing power consumption.

Another important factor is convenience. Smart home systems allow users to control appliances remotely using mobile devices or voice commands. This eliminates the need for manual operation and makes daily tasks easier and more efficient.

Safety and security are also major concerns in modern homes. Smart systems can include features such as automatic lighting, fire detection, and intrusion alerts, providing an additional layer of protection.

Furthermore, the increasing number of elderly and physically challenged individuals highlights the importance of automation. Voice-controlled systems enable such individuals to operate appliances easily without physical effort.

In addition, the concept of smart cities and digital infrastructure has increased the demand for intelligent home systems. These systems play a crucial role in building connected and sustainable environments.

Therefore, the development of smart home automation systems is essential to meet the evolving needs of modern society.

## **1.5 Problem Statement**

Despite the availability of various home automation technologies, several challenges still exist in traditional and existing systems. Many conventional systems rely on manual control, which leads to inefficiency and inconvenience. Users must physically operate switches, which can be time-consuming and impractical in certain situations.

Existing automation systems often suffer from high cost and complexity, making them inaccessible to a large section of users. Additionally, some systems are limited to a single mode of control, such as only Wi-Fi or only Bluetooth, reducing flexibility.

Another major issue is inefficient energy management. Appliances may remain ON even when not required, leading to unnecessary power consumption. Moreover, many systems do not utilize real-time environmental data for intelligent decision-making.

There is also a lack of integration between automatic and manual control. Users may not have the option to override automatic settings when needed.

Therefore, there is a need for a smart home system that:

- Provides both automatic and manual control
- Is cost-effective and easy to implement
- Utilizes real-time sensor data
- Supports multiple communication methods
- Improves energy efficiency and convenience

This project aims to address these challenges by developing an IoT-based smart home automation system using ESP32.

## **1.6 Objectives of the Project**

The main objectives of this project are:

- To design and develop a smart home automation system using ESP32
- To monitor environmental conditions such as temperature and light intensity
- To control appliances automatically based on sensor data
- To provide manual control using Wi-Fi and Bluetooth
- To implement voice-based control for user convenience
- To improve energy efficiency and reduce power consumption
- To develop a low-cost and user-friendly system

## **1.7 Scope of the Project**

The scope of this project focuses on the development and implementation of a basic smart home automation system using IoT technology. The system is designed to control appliances such as lights and fans based on environmental conditions and user commands.

The project includes:

- Integration of sensors (DHT11 and LDR)
- Use of ESP32 for processing and communication
- Implementation of relay-based appliance control
- Development of a web interface for monitoring and control
- Bluetooth-based command system

Although the current system is designed for basic home automation, it can be extended in the future to include additional features such as:

- Mobile application control
- Cloud-based monitoring
- AI-based automation
- Integration with smart assistants

The project serves as a foundation for advanced IoT-based systems and provides valuable insights into embedded system design and wireless communication.

## CHAPTER 2

### LITERATURE SURVEY

#### 2.1 Literature Review

The development of smart home automation systems has attracted significant attention from researchers in recent years, particularly with the emergence of IoT technologies. Various studies have explored different approaches, communication methods, and control mechanisms.

[1] **H. Kim, J. Lee, and K. Park (2017)** proposed an IoT-based smart home system that integrates cloud computing for remote monitoring and control. The system utilized sensors and mobile applications to provide real-time data access. The study highlighted the importance of cloud connectivity in enhancing system accessibility, but also pointed out dependency on continuous internet connectivity.

[2] **R. Piyare and M. Tazil (2011)** developed a Bluetooth-based home automation system using a microcontroller. The system allowed users to control appliances using a mobile device within a limited range. While the system was simple and cost-effective, it lacked scalability and remote access capabilities.

[3] **D. Pavithra and R. Balakrishnan (2015)** introduced a GSM-based home automation system that enabled users to control appliances through SMS commands. The system provided long-distance control but suffered from delays in message transmission and increased operational costs.

[4] **S. Madakam, R. Ramaswamy, and S. Tripathi (2015)** discussed the role of IoT in smart environments, including home automation. The study emphasized the importance of interconnected devices and real-time data processing in improving system efficiency and user convenience.

[5] **Espressif Systems (ESP32 Documentation)** demonstrated the application of ESP32 in IoT-based systems. The ESP32 microcontroller was identified as a powerful and cost-effective solution due to its integrated Wi-Fi and Bluetooth capabilities, making it suitable for smart home applications.

[6] **A. Alheraish (2004)** proposed a home automation system using microcontrollers and sensors. The system provided basic automation features but lacked wireless communication and scalability.

[7] **T. Perumal et al. (2008)** developed a smart home system using wireless sensor networks. The system improved communication between devices but required complex network configurations. From the above studies, it is evident that various technologies such as Bluetooth, GSM, and IoT have been used in home automation systems. However, each approach has its own limitations in terms of cost, scalability, and flexibility.

## **2.2 Limitations of Conventional Systems**

Although significant advancements have been made in the field of home automation, conventional systems still suffer from several limitations that restrict their efficiency, scalability, and practical usability. These limitations become more prominent when such systems are implemented in real-world environments, especially at a larger scale.

### **1. High Installation and Maintenance Cost**

Many traditional and early smart home systems require specialized hardware, proprietary modules, and professional installation. Systems based on ZigBee, KNX, or wired automation involve additional infrastructure costs such as gateways, controllers, and dedicated wiring.

Moreover, maintenance of these systems can be expensive, as they often require technical expertise for troubleshooting and upgrades. This makes such systems less accessible to middle-and low-income users.

### **2. Complexity in System Design and Configuration**

Conventional systems often involve complex architectures with multiple components and communication protocols. Configuring these systems requires in-depth technical knowledge of hardware interfaces, networking, and programming.

For example:

- Setting up GSM modules requires SIM configuration
- ZigBee networks require node pairing and addressing
- Wired systems require structured cabling

This complexity makes the system difficult for beginners and limits its widespread adoption.

### **3. Limited Communication Range and Flexibility**

Systems based on Bluetooth or infrared (IR) communication suffer from limited range and line-of-sight constraints. Bluetooth-based systems typically operate within a range of 10–20 meters, making them unsuitable for large homes or remote access.

Similarly, IR-based systems require direct alignment between transmitter and receiver, which reduces flexibility and usability.

#### **4. Lack of Remote Accessibility**

Many conventional systems do not support internet-based control, restricting users to operate appliances only within a local environment. For example:

- Bluetooth systems require proximity
- IR systems require direct access

This limitation prevents users from controlling devices remotely, reducing the overall convenience of the system.

#### **5. Dependency on Single Communication Technology**

Traditional systems often rely on a single communication method such as GSM, Bluetooth, or wired connections. This creates several issues:

- If the communication channel fails, the entire system becomes non-functional
- Lack of backup communication reduces reliability
- No flexibility to switch between control modes

Modern systems require hybrid communication (e.g., Wi-Fi + Bluetooth), which is missing in conventional designs.

#### **6. Inefficient Energy Management**

One of the major drawbacks of conventional systems is the lack of intelligent energy management. Appliances are often operated manually without considering environmental conditions.

For example:

- Lights remain ON even in daylight
- Fans operate continuously regardless of temperature

This leads to excessive energy consumption and increased electricity costs. Conventional systems do not utilize sensor-based automation effectively.

#### **7. Absence of Real-Time Monitoring**

Most traditional systems do not provide real-time monitoring of environmental parameters such as temperature, humidity, or light intensity. Users are unable to track system performance or make informed decisions based on data.

Without real-time feedback:

- System efficiency cannot be optimized
- Fault detection becomes difficult

## **CHAPTER 3**

### **SYSTEM DESIGN AND THEORY**

#### **3.1 Introduction**

The design of a smart home automation system involves the integration of sensors, microcontrollers, communication modules, and control devices to achieve intelligent operation of household appliances. In this project, the system is designed using the ESP32 microcontroller, which acts as the central processing unit.

The primary objective of the system design is to develop an efficient and reliable automation mechanism that can monitor environmental conditions and control appliances accordingly. The system incorporates both automatic and manual control modes, ensuring flexibility and user convenience.

The design focuses on:

- Sensor data acquisition
- Data processing using ESP32
- Decision-making based on predefined logic
- Control of appliances using relays
- Communication through Wi-Fi and Bluetooth

#### **3.2 System Architecture**

The system architecture represents the overall structure and interaction between different components of the smart home system. It consists of four main sections:

1. **Input Section (Sensors)**
2. **Processing Section (ESP32 Controller)**
3. **Control Section (Relay Module)**
4. **Output Section (Appliances)**

##### **Working Flow:**

- Sensors collect environmental data
- ESP32 processes the data
- Control logic is applied
- Relay modules switch appliances ON/OFF
- User interacts via Wi-Fi or Bluetooth

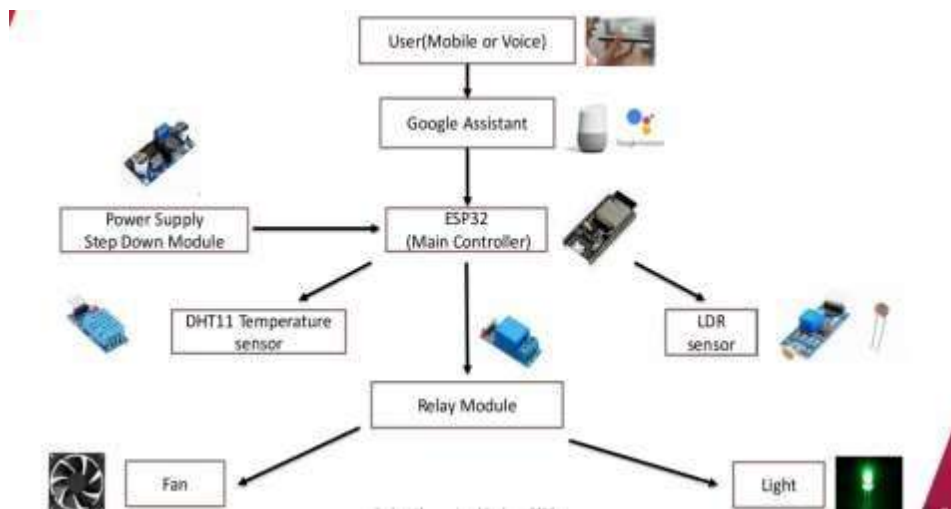
This architecture ensures efficient communication between hardware and software components.

### 3.3 Block Diagram Description

The block diagram of the system includes the following components:

- **DHT11 Sensor** → **ESP32** (Temperature & Humidity input)
- **LDR Sensor** → **ESP32** (Light detection input)
- **ESP32** → **Relay Module** (Control signals)
- **Relay Module** → **Appliances** (Fan & Light)
- **ESP32** → **Web Server / Bluetooth** (User interface)

The ESP32 acts as the central node connecting all modules and enabling communication.



*Fig 3.3 Block Diagram Description*

### 3.4 Working Principle of the System

The working principle of the proposed system is based on real-time sensing, data processing, and automated decision-making.

#### Step-by-Step Operation:

1. The system is powered ON and initializes all components.
2. ESP32 connects to Wi-Fi and enables Bluetooth communication.
3. Sensors start collecting environmental data.
4. The DHT11 sensor measures temperature and humidity.
5. The LDR sensor detects light intensity (dark/bright).
6. The ESP32 processes sensor data using predefined logic.
7. Based on the conditions:
  - Fan is turned ON/OFF
  - Light is turned ON/OFF

8. Users can also control appliances manually via:

- Web interface (Wi-Fi)
- Bluetooth commands

Thus, the system ensures both automatic and manual control.

### 3.5 Components Description

#### 3.5.1 ESP32 Microcontroller

The ESP32 is the core component of the system. It performs the following functions:

- Reads sensor data
- Processes control logic
- Communicates via Wi-Fi and Bluetooth
- Controls relay outputs

Key features:

- Dual-core processor
- Built-in Wi-Fi and Bluetooth
- Multiple GPIO pins
- Low power consumption



*Fig 3.5.1 ESP32 Microcontroller*

#### 3.5.2 DHT11 Sensor

The DHT11 sensor is used to measure temperature and humidity.

##### Working Principle:

- Uses a thermistor for temperature measurement
- Uses capacitive sensing for humidity
- Outputs digital signal to ESP32

##### Specifications:

- Temperature range: 0°C to 50°C
- Humidity range: 20% to 90%



*Fig 3.5.2 DHT11 Sensor*

### **3.5.3 LDR Sensor (Light Dependent Resistor)**

The LDR sensor is used to detect light intensity.

#### **Working Principle:**

- Resistance decreases with increase in light
- Resistance increases in darkness
- Provides digital signal (HIGH/LOW)

This helps in automatic light control.



*Fig 3.5.3 LDR Sensor*

### 3.5.4 Relay Module

The relay acts as an electrically operated switch used to control appliances.

#### Working Principle:

- Operates using electromagnetic induction
- A low voltage signal from ESP32 controls high voltage devices

#### Function:

- Turns ON/OFF fan and light



*Fig 3.5.4 Relay Module*

### 3.5.5 Power Supply

A regulated power supply is used to provide stable voltage to the system. The ESP32 operates at 3.3V, while relays may require 5V.

## 3.6 Automation Logic

The system uses simple threshold-based logic for automation.

#### Temperature Control Logic

- If Temperature  $> 35^{\circ}\text{C}$   $\rightarrow$  Fan ON
- If Temperature  $< 34^{\circ}\text{C}$   $\rightarrow$  Fan OFF

(This creates hysteresis to avoid frequent switching)

#### Light Control Logic

- If Dark (LDR HIGH)  $\rightarrow$  Light ON
- If Bright (LDR LOW)  $\rightarrow$  Light OFF

This ensures efficient energy usage.

### **3.7 Communication Mechanisms**

The system supports dual communication methods:

#### **Wi-Fi Communication**

- ESP32 hosts a web server
- User accesses system via browser
- Displays sensor data and control options

#### **Bluetooth Communication**

- Uses Bluetooth Serial
- Accepts commands from mobile devices
- Example commands:
  - FAN ON
  - LIGHT OFF
  - STATUS

This dual-mode communication increases system flexibility.

### **3.8 Web-Based Control System**

The ESP32 hosts a local web server that provides a user interface.

#### **Features:**

- Displays real-time data
- Allows manual control
- Auto-refreshes periodically

#### **Control Options:**

- Fan ON / AUTO
- Light ON / AUTO

This makes the system user-friendly and interactive.

### **3.9 System Design Considerations**

Several factors are considered while designing the system:

#### **1. Cost Efficiency**

Low-cost components are used to make the system affordable.

#### **2. Reliability**

Stable communication and sensor readings are ensured.

#### **3. Scalability**

Additional devices can be added easily.

#### **4. Power Efficiency**

Automation reduces unnecessary energy consumption.

#### **5. User Convenience**

Multiple control methods improve usability.

### **3.10 Advantages of Proposed Design**

- Simple and easy to implement
- Supports both automatic and manual control
- Low power consumption
- Cost-effective solution
- Flexible and scalable system
- Real-time monitoring and control

## CHAPTER 4

### PROPOSED METHODOLOGY

#### 4.1 Introduction

The proposed methodology describes the systematic approach followed to design, develop, and implement the IoT-based voice-enabled smart home automation system. It focuses on integrating hardware and software components to achieve efficient automation and control of household appliances.

The methodology combines:

- Sensor-based data acquisition
- Embedded system processing (ESP32)
- Wireless communication (Wi-Fi and Bluetooth)
- Automation logic and control mechanisms

The objective is to develop a system that is **efficient, reliable, user-friendly, and cost-effective**, while overcoming the limitations of conventional systems discussed in Chapter 2.

#### 4.2 System Design Approach

The system is designed using a modular approach, where each component performs a specific function. The overall system is divided into the following modules:

##### 1. Sensing Module

- Collects environmental data
- Uses DHT11 and LDR sensors

##### 2. Processing Module

- ESP32 processes sensor data
- Executes control algorithms

##### 3. Communication Module

- Wi-Fi for web-based control
- Bluetooth for local/voice control

##### 4. Control Module

- Relay switches appliances ON/OFF

##### 5. User Interface Module

- Web dashboard
- Bluetooth command interface

This modular design ensures easy implementation, debugging, and future expansion.

### **4.3 System Flow Description**

The working of the system follows a structured flow:

1. System initialization
2. Sensor data acquisition
3. Data processing by ESP32
4. Decision-making using automation logic
5. Control signals sent to relay module
6. Appliances are switched accordingly
7. User commands are processed (if any)
8. System repeats continuously

This continuous loop ensures real-time monitoring and control.

### **4.4 Flow Diagram Explanation**

Although the diagram is not shown here, the methodology can be explained step-by-step:

#### **Step 1: Start**

The system begins operation when power is supplied to the ESP32.

#### **Step 2: Initialization**

- Initialize sensors (DHT11, LDR)
- Configure GPIO pins
- Start Wi-Fi and Bluetooth

#### **Step 3: Data Acquisition**

- Read temperature and humidity from DHT11
- Read light status from LDR

#### **Step 4: Decision Making**

- Compare sensor values with predefined thresholds
- Determine whether to turn appliances ON or OFF

#### **Step 5: Appliance Control**

- Send signals to relay module
- Control fan and light

#### **Step 6: User Input Check**

- Check for commands from web interface
- Check for Bluetooth commands

### **Step 7: Override Logic**

- If user command exists → manual control
- Else → automatic control

### **Step 8: Repeat Loop**

The system continuously monitors and updates.

## **4.5 Automation Method**

The system uses **rule-based automation**, where predefined conditions control the appliances.

### **Temperature-Based Automation**

- If temperature exceeds threshold → fan ON
- If temperature drops → fan OFF

This prevents overheating and improves comfort.

### **Light-Based Automation**

- If environment is dark → light ON
- If environment is bright → light OFF

This ensures energy-efficient lighting.

### **Hysteresis Control**

A small difference between ON and OFF thresholds prevents frequent switching of appliances, increasing system stability.

#### **4.6 Inset Control: Manual Override Mechanism**

One of the key features of the proposed system is the **manual override mechanism**, which allows users to control appliances manually even when automatic mode is active.

##### **Types of Manual Control:**

##### **1. Web-Based Control**

- User accesses local web server
- Controls appliances via buttons

##### **2. Bluetooth Control**

- Commands sent via mobile device
- Example commands:
  - FAN ON
  - LIGHT OFF

This dual control system increases flexibility and user convenience.

#### **4.7 Communication Methodology**

The system uses two communication methods:

##### **Wi-Fi Communication**

- ESP32 connects to local network
- Hosts web server
- Enables real-time monitoring

##### **Bluetooth Communication**

- Uses Bluetooth Serial
- Allows control without internet

##### **Advantages of Dual Communication**

- Increased reliability
- Backup control option
- Better user experience

#### **4.8 Comparison with Conventional Methods**

| <b>Feature</b>    | <b>Conventional Systems</b> | <b>Proposed System</b> |
|-------------------|-----------------------------|------------------------|
| Control Type      | Manual only                 | Automatic + Manual     |
| Communication     | Single mode                 | Dual mode              |
| Cost              | High                        | Low                    |
| Flexibility       | Limited                     | High                   |
| Energy Efficiency | Low                         | High                   |

## **CHAPTER 5**

### **IMPLEMENTATION**

#### **5.1 Introduction**

The implementation phase is a crucial part of the project, where the theoretical design and methodology are converted into a working system. This chapter describes the practical realization of the IoT-based voice-enabled smart home automation system using hardware components and software tools.

The implementation involves:

- Hardware setup and connections
- Software development using Arduino IDE
- Programming of ESP32 microcontroller
- Integration of sensors and actuators
- Testing and validation of system performance

The goal is to ensure that the system operates efficiently under real-time conditions.

#### **5.2 Hardware Implementation**

**The hardware implementation involves assembling all components and establishing proper connections.**

Components Used:

- ESP32 Development Board
- DHT11 Temperature and Humidity Sensor
- LDR Sensor Module
- 2-Channel Relay Module
- Fan (Load)
- Light (Load)
- Power Supply
- Connecting Wires

### **5.2.1 ESP32 Setup**

The ESP32 acts as the central controller of the system. It is powered using a USB cable or regulated power supply. GPIO pins are configured for sensor input and relay output.

Functions performed:

- Reads sensor data
- Executes control logic
- Handles Wi-Fi and Bluetooth communication
- Controls appliances

### **5.2.2 Sensor**

#### **Connections**

DHT11 Sensor

- Connected to GPIO pin (e.g., GPIO13)
- Provides temperature and humidity data

LDR Sensor

- Connected to GPIO pin (e.g., GPIO27)
- Detects light intensity

### **5.2.2 Relay Module Connections**

The relay module is used to control high-voltage appliances.

- Fan connected to Relay Channel 1
- Light connected to Relay Channel 2
- Relay inputs connected to ESP32 GPIO pins

Note: Relay operates in active-low mode.

### **5.2.3 Power Supply Setup**

- ESP32 operates at 3.3V
- Relay module may require 5V
- Stable power supply ensures proper functioning

### 5.3 Software Implementation

The software implementation is carried out using the Arduino IDE, which provides an easy environment for programming the ESP32.

#### 5.3.1 Development Environment

- Arduino IDE
- ESP32 Board Package
- Required Libraries:
  - WiFi.h
  - WebServer.h
  - BluetoothSerial.h
  - DHT.h

#### 5.3.2 Program Structure

The program is divided into the following sections:

##### 1. Library Inclusion

Includes necessary libraries for Wi-Fi, Bluetooth, and sensors.

##### 2. Global Configuration

- Define pin numbers
- Set Wi-Fi credentials
- Initialize variables

##### 3. Setup Function

- Initialize serial communication
- Configure GPIO pins
- Start sensors
- Enable Wi-Fi and Bluetooth
- Start web server

##### 4. Loop Function

- Read sensor data
- Execute automation logic
- Check user commands
- Control appliances

## 5.4 Wi-Fi Configuration and Web Server

The ESP32 connects to a Wi-Fi network and hosts a local web server.

Features of Web Server:

- Displays real-time sensor data
- Provides control buttons
- Accessible via browser

Working:

- ESP32 gets IP address
- User enters IP in browser
- Dashboard is displayed

## 5.5 Bluetooth Implementation

Bluetooth communication allows local control without internet.

Working:

- ESP32 acts as Bluetooth device
- User connects via mobile
- Commands are sent as text

Example Commands:

- FAN ON
- FAN OFF
- LIGHT ON
- LIGHT OFF

Note:

Commands must be processed in uppercase for proper matching.

## 5.6 Sensor Data Processing

The ESP32 continuously reads sensor values:

- Temperature and humidity from DHT11
- Light status from LDR

If sensor reading fails:

- Default values are used
- Error handling ensures stability

## **5.7 Implementation Challenges**

During implementation, some challenges may arise:

- Sensor inaccuracies (DHT11 limitations)
- Wi-Fi connectivity issues
- Bluetooth command mismatch
- Relay noise and switching delays

These issues are addressed through proper coding and configuration.

## **5.8 Advantages of Implementation**

- Simple hardware setup
- Easy programming using Arduino IDE
- Low-cost components
- Reliable communication
- Real-time operation

## CHAPTER 6

### RESULTS AND DISCUSSION

#### 6.1 Introduction

This chapter presents the results obtained from the implementation of the IoT-based voice-enabled smart home automation system. The performance of the system is evaluated based on various parameters such as temperature control, light automation, communication efficiency, and overall system reliability.

The results are analyzed to determine whether the system meets the objectives defined in earlier chapters. The discussion also highlights the effectiveness of automation logic, communication methods, and system response under different operating conditions

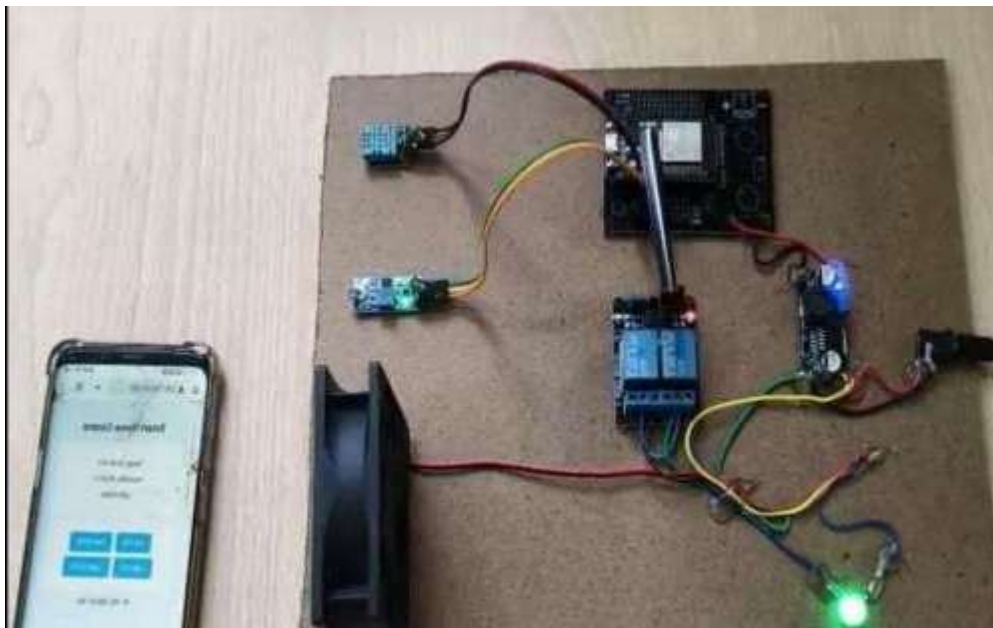


Fig 6.1 Result

#### 6.2 Temperature-Based Automation Analysis

The DHT11 sensor is used to monitor temperature continuously, and the system controls the fan based on predefined threshold values.

##### Observed Behavior:

- When temperature exceeds **35°C**, the fan turns ON automatically
- When temperature drops below **34°C**, the fan turns OFF

##### Analysis:

The system successfully responds to temperature variations and controls the fan accordingly. The

use of two threshold values creates a **hysteresis effect**, which prevents frequent switching of the relay near the threshold point.

**Result:**

- Efficient temperature-based control achieved
- Stable operation without rapid ON/OFF switching

### **6.3 Light-Based Automation Analysis**

The LDR sensor is used to detect ambient light conditions and control lighting automatically.

**Observed Behavior:**

- In dark conditions → Light turns ON
- In bright conditions → Light turns OFF

**Analysis:**

The system accurately detects changes in light intensity and responds appropriately. Since the LDR is used in digital mode, the system operates based on two states: DARK and BRIGHT.

**Result:**

- Effective automatic lighting system
- Reduced energy wastage

### **6.4 Web-Based Control Analysis**

The ESP32 hosts a local web server that allows users to monitor and control the system through a web browser.

**Observed Features:**

- Displays real-time temperature and humidity
- Shows light status (Dark/Bright)
- Provides control buttons for appliances
- Auto-refreshes periodically

**Performance:**

- Fast response time
- Easy accessibility within local network
- User-friendly interface

**Result:**

- Successful implementation of web-based control
- Reliable real-time monitoring

## **6.5 Bluetooth Control Analysis**

Bluetooth communication allows users to control appliances using mobile devices without internet connectivity.

### **Observed Behavior:**

- Commands like “FAN ON”, “LIGHT OFF” control appliances
- System responds immediately

### **Issue Identified:**

- Command mismatch occurs if case sensitivity is not handled properly

### **Solution:**

- Convert input commands to uppercase

### **Result:**

- Effective offline control system
- Improved flexibility

## 6.6 Web-Based Control Analysis

The ESP32 hosts a local web server that allows users to monitor and control the system through a web browser.

### Observed Features:

- Displays real-time temperature and humidity
- Shows light status (Dark/Bright)
- Provides control buttons for appliances
- Auto-refreshes periodically

### Performance:

- Fast response time
- Easy accessibility within local network
- User-friendly interface

### Result:

- Successful implementation of web-based control
- Reliable real-time monitoring

## 6.7 Bluetooth Control Analysis

Bluetooth communication allows users to control appliances using mobile devices without internet connectivity.

### Observed Behavior:

- Commands like “FAN ON”, “LIGHT OFF” control appliances
- System responds immediately

### Issue Identified:

- Command mismatch occurs if case sensitivity is not handled properly

### Solution:

- Convert input commands to uppercase

### Result:

- Effective offline control system
- Improved flexibility

## 6.8 System Response and Performance

### Response Time:

- Sensor reading: Instant
- Relay switching: Within milliseconds
- Web response: Fast within local network

### Reliability:

- System operates continuously without failure
- Stable communication through Wi-Fi and Bluetooth

### Accuracy:

- DHT11 provides acceptable accuracy for basic applications
- LDR provides reliable light detection

## 6.9 Power Efficiency Analysis

One of the main objectives of the system is to improve energy efficiency.

### Observations:

- Appliances operate only when required
- Automatic control reduces unnecessary usage

### Example:

- Light turns OFF during daytime
- Fan turns OFF when temperature is low

### Result:

- Significant reduction in power consumption
- Improved energy management

## 6.10 Comparison with Manual System

| Parameter         | Manual System | Proposed System    |
|-------------------|---------------|--------------------|
| Control           | Manual        | Automatic + Manual |
| Energy Efficiency | Low           | High               |
| Convenience       | Low           | High               |
| Remote Access     | Not available | Available          |
| Response Time     | Slow          | Fast               |

## CHAPTER 7

### CONCLUSION AND FUTURE SCOPE

#### 7.1 Conclusion

In this project, an **IoT-based voice-enabled smart home automation system** has been successfully designed, implemented, and tested using the ESP32 microcontroller. The system integrates sensing, processing, communication, and control mechanisms to automate household appliances efficiently.

The project began with the identification of limitations in conventional home automation systems, such as high cost, lack of flexibility, and inefficient energy usage. To address these challenges, a smart automation system was developed using ESP32, which provides built-in Wi-Fi and Bluetooth capabilities, making it suitable for IoT applications.

The system utilizes:

- **DHT11 sensor** for temperature and humidity monitoring
- **LDR sensor** for light detection
- **Relay module** for controlling appliances such as fan and light

The automation logic was designed to control appliances based on environmental conditions. The fan operates according to temperature thresholds, while the lighting system responds to ambient light conditions. This ensures efficient energy utilization and reduces manual effort.

In addition to automatic control, the system provides **manual control through both Wi-Fi and Bluetooth communication**. The ESP32 hosts a local web server that allows users to monitor real-time data and control appliances using a browser interface. Bluetooth communication enables control without internet access, enhancing system flexibility.

The results obtained from the implementation demonstrate that:

- The system operates reliably under different conditions
- Automation logic performs accurately
- Communication is stable and responsive
- Energy efficiency is improved

The project successfully achieves its objectives of developing a **low-cost, efficient, and user-friendly smart home automation system**. It also demonstrates the practical application of IoT technology in real-world scenarios.

Overall, the system provides a strong foundation for advanced smart home solutions and highlights the potential of embedded systems in modern automation

## **7.2 Future Scope**

Although the proposed system performs effectively, there are several opportunities for further improvement and enhancement. Future developments can focus on improving system performance, scalability, and functionality.

### **1. Mobile Application Integration**

A dedicated mobile application can be developed to provide a more user-friendly interface. This would allow users to control appliances more efficiently compared to a basic web interface.

### **2. Cloud-Based Monitoring and Control**

The system can be integrated with cloud platforms such as AWS, Firebase, or IoT dashboards.

This would enable:

- Remote access from anywhere in the world
- Data storage and analytics
- Real-time monitoring

### **3. Voice Assistant Integration**

The system can be enhanced by integrating with voice assistants such as:

- Google Assistant
- Amazon Alexa

This would enable complete voice-based control, improving accessibility and user convenience.

### **4. Advanced Sensors Integration**

More advanced sensors can be added to improve system functionality, such as:

- Motion sensors for automatic lighting
- Gas sensors for safety
- Smoke sensors for fire detection

### **5. Improved User Interface**

The web interface can be upgraded using:

- Real-time updates (AJAX/WebSocket)
- Graphical dashboards
- Interactive controls

This would enhance user experience and system usability.

## CHAPTER 8

### SUSTAINABLE DEVELOPMENT GOALS (SDGs)

#### 8.1 Introduction

Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations to promote sustainable growth, innovation, and improved quality of life. Engineering projects, especially in IoT and automation, play an important role in achieving these goals.

The proposed **IoT-Based Smart Home Automation System** contributes to sustainability by improving energy efficiency and enabling intelligent control of household appliances.

#### 8.2 Relevant SDGs

##### **SDG 9: Industry, Innovation, and Infrastructure**

The project promotes innovation by using IoT technology and ESP32 microcontroller for smart automation. It supports the development of modern and efficient technological solutions.

##### **SDG 11: Sustainable Cities and Communities**

The system contributes to smart home development, which is an essential part of smart cities. It improves convenience, safety, and efficient resource usage in residential environments.

##### **SDG 7: Affordable and Clean Energy**

The system reduces energy wastage by automatically controlling appliances based on environmental conditions. This leads to efficient use of electricity and supports sustainable energy practices.

##### **SDG 4: Quality Education**

The project enhances practical knowledge in IoT, embedded systems, and automation. It helps students understand real-world applications of engineering concepts.

#### 8.3 Impact of the Project

- Reduces energy consumption
- Improves user convenience
- Supports smart living environments
- Encourages technological innovation

#### **8.4 Summary**

The proposed system aligns with multiple SDGs by promoting energy efficiency, innovation, and smart infrastructure. It demonstrates how IoT-based solutions can contribute to sustainable development.

## REFERENCES

- [1] Espressif Systems, *ESP32 Technical Reference Manual*, Espressif Systems, 2023.
- [2] Arduino, “Arduino IDE Documentation,” [Online]. Available: <https://www.arduino.cc/en/software>
- [3] R. Piyare and M. Tazil, “Bluetooth Based Home Automation System Using Cell Phone,” *IEEE 15th International Symposium on Consumer Electronics*, 2011.
- [4] D. Pavithra and R. Balakrishnan, “IoT based Monitoring and Control System for Home Automation,” *IEEE Global Conference on Communication Technologies*, 2015.
- [5] S. Madakam, R. Ramaswamy, and S. Tripathi, “Internet of Things (IoT): A Literature Review,” *Journal of Computer and Communications*, vol. 3, no. 5, pp. 164–173, 2015.
- [6] A. Alheraish, “Design and Implementation of Home Automation System,” *IEEE Transactions on Consumer Electronics*, vol. 50, no. 4, pp. 1087–1092, 2004.
- [7] T. Perumal, M. Ramli, and C. Y. Leong, “Interoperability of Smart Home Environment Using Web Services,” *IEEE International Conference on Intelligent and Advanced Systems*, 2008.
- [8] Aosong Electronics, *DHT11 Temperature and Humidity Sensor Datasheet*, 2022.
- [9] Electronics Tutorials, “Relay Working Principle,” [Online]. Available: <https://www.electronics-tutorials.ws>
- [10] Last Minute Engineers, “ESP32 Web Server Tutorial,” [Online]. Available: <https://lastminuteengineers.com>